

PROJECT REPORT

AI Based Career Recommendation System

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1. Abstract

This project is about making a simple system that can suggest a career based on a student's skills, interests and academic information. The user enters details such as programming knowledge, mathematics, communication, problem solving and interest in different fields. A machine learning model uses these details and predicts a suitable career. A Streamlit application is used so that the system can be used easily in a browser.

2. Introduction

Choosing a career can be confusing for students because everyone has different skills and interests. The main idea of this project is to use machine learning to give a basic career suggestion from the information provided by the student. This project is mainly developed to understand how a machine learning model can be used in a practical application.

3. Problem Statement

Many students are not sure which technology career is suitable for them. A simple system is needed that can take information about a student's skills and interests and give a possible career recommendation.

4. Objectives

- To build a simple machine learning based career recommendation system.
- To use student skills and interests as input.
- To train a classification model for career prediction.
- To create a simple interface using Streamlit.
- To understand the basic steps of a machine learning project.

5. Proposed Solution

The system takes student information as input. The data is prepared and passed to a trained Random Forest Classifier. The model predicts a career category and the result is shown in the application. The recommendation is only a starting point and should not be treated as a final career decision.

6. Working of the System

Student Details → Data Preprocessing → Model Training → Career Prediction → Result

The student first enters the required information. The trained model then uses the input values to predict the career category.

7. Technologies Used

Python is used as the main programming language. Pandas and NumPy are used for handling data. Matplotlib can be used for basic data visualization. Scikit-learn is used for machine learning. Joblib can be used to save the trained model. Streamlit is used to create the user

interface. Git and GitHub are used for version control.

8. Dataset

The dataset contains information about student skills, interests and academic performance. Some example columns are Python, Java, C++, Mathematics, Communication, Problem Solving, AI/ML Interest, Web Development Interest, Data Science Interest and Academic Performance. The Career column is used as the target value.

9. Machine Learning Model

The project uses the Random Forest Classifier. It works by combining the results of multiple decision trees. The model is used as a classification model because the output is a career category. The actual performance depends on the dataset and the way the data is prepared.

10. Implementation

The project is divided into a few simple parts. First, the dataset is loaded and prepared. Next, the features and target are separated and the data is divided into training and testing parts. The Random Forest model is trained using the training data. After testing the model, it can be saved using Joblib. Finally, the Streamlit application takes user input and uses the saved model to make a prediction.

11. How to Run the Project

1. Install Python 3.9 or above.
2. Download or clone the project repository.
3. Open the project folder in the terminal.
4. Create a virtual environment if required.
5. Install the libraries using requirements.txt.
6. Train the model if the model file is not already present.
7. Run: `streamlit run app.py`
8. Open the local address shown by Streamlit in the browser.

12. Testing and Evaluation

The model can be checked using accuracy, precision, recall and F1-score. A confusion matrix can also be used to understand the predictions for different career categories. The final accuracy value should be filled in after running the actual project. No fixed accuracy value is assumed in this report.

13. Expected Result

After entering the required information, the application displays a predicted career. Possible results can include AI/ML Engineer, Software Developer, Data Scientist, Data Analyst, Web Developer and Cybersecurity Analyst, depending on the dataset.

14. Advantages

- Easy to use.
- Uses machine learning for prediction.
- Can be improved by adding more data.
- Gives practical experience with a complete ML workflow.
- Can be extended with more career options.

15. Limitations

The result depends on the training dataset and the information entered by the user. A machine learning model cannot understand every personal factor involved in choosing a career. Because of this, the result should be used only as general guidance.

16. Future Scope

The project can be improved by adding resume analysis, skill gap analysis, learning recommendations and job suggestions. More career categories can also be added. In the future, NLP, a database, user accounts and online deployment can be included.

17. Conclusion

The AI Based Career Recommendation System is a simple application of machine learning to a student-related problem. It takes information about skills and interests and gives a possible career prediction. Through this project, concepts such as data preprocessing, classification, model evaluation and Streamlit application development can be practiced.

18. Repository

GitHub Repository: <YOUR_GITHUB_REPOSITORY_URL>

The repository should contain the source code, dataset, model or training script, requirements.txt and README.md.

19. References

1. Python Documentation
2. Pandas Documentation
3. NumPy Documentation
4. Scikit-learn Documentation
5. Streamlit Documentation